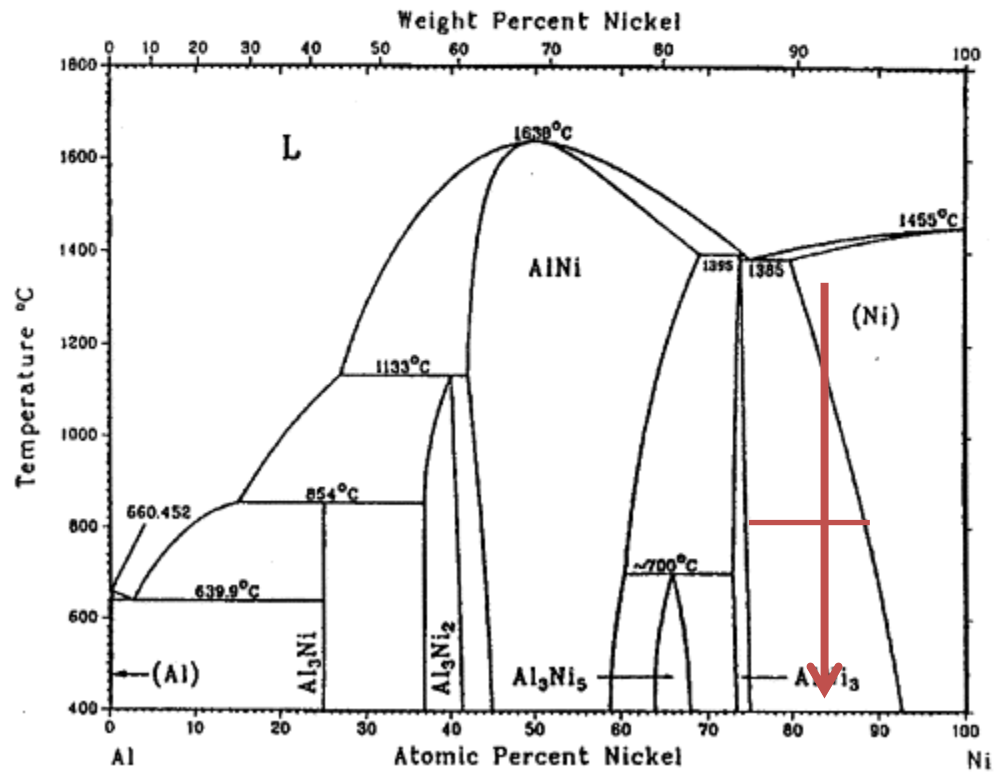
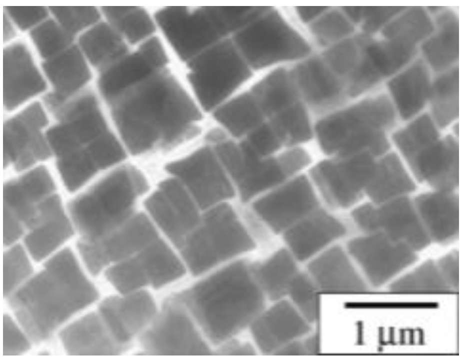
2. Alloys: High temperature alloys
(Lecture slides together with 3. Failure: Creep)

1. Ni-Al system as a basis of superalloys



Understand the heat treatment according to the Al-Ni phase diagram:
Target is a uniform and fine grain of up to 60% Ni_3Al in a (Ni) matrix.
Solutionise in Ni field to dissolve all Al into solution; Then quench to create Super-Saturated Solid Solution; aged at 800 °C for 8 hours to control the precipitation of Ni3Al throughout the Ni matrix (as in image below).



The peak-aged Ni-Ni3Al has a good strength and thermal stability.
Strength: Even though the interface is coherent, dislocation find it difficult to move through the Ni3Al phase because Ni3Al is ordered.
The passage of a dislocation through Ni3Al disrupts the order. Dislocations must pass through in pairs: one disrupts and the other restores the order, which leads to significant strengthening.

Thermal stability: This is good for minimizing ____cousening____ (the decrease in interfacial area with time).

2. Real Ni superalloy compositions:

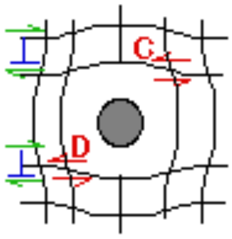
Current turbine blade technology is based on multi-component Ni-based superalloys containing only about 60% Ni.

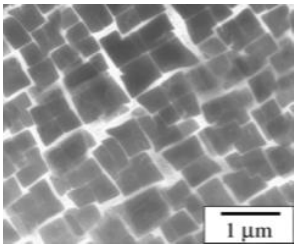
The compositions include:

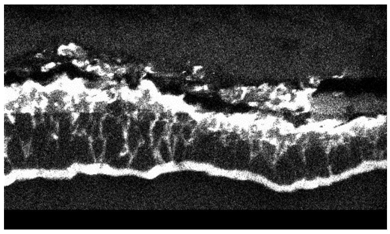
soild solution strengtheners e.g. about 10% each of Co, _W_, Cr.

_precipitates__ formers e.g. _AL_, Ti, Mo, Ta, C.

And __oxidation__ protection, e.g. _Cr_.







3. The evolution of turbine blade materials against creep.

<u>Design against dislocation creep</u>

1a Choose a material with a high melting point (minimise T/T_m)

1b Maximise obstacles to dislocation motion that are stable at operation T

1c Choose an alloy where the majority phase has a high lattice resistance

<u>Design against diffusion creep</u>

2a Choose a material with a high melting point (minimise T/T_m)

2b Maximise the grain size to give long diffusion distances and little gb diffusion.

2c Choose an alloy with precipitates at gb's that are stable at operation T to impede gb sliding

In general, Ni3Al is a good barrier hindering _dislocation_ motion at elevated T supporting against __dislocation__ creep (__1b__in the chart above).

- For equiaxed turbine blade (early generation): when _polycrstalline__ blades were used, precipitates were required at _grain boundaries__ to reduce grain boundary _sliding__, e.g. carbides $M_{23}C_6$ (M= Cr, Mo etc) (__2c__in the chart above). But __diffusion creep____ is still a problem at high temperature
- For columnar turbine blade: The columnar grain will __muximise__ the grain size to give long diffusion __distance___ and little grain boundary diffusion. Align grains in __louding__ direction to resist diffusion creep

(2b in the chart above). It has also carbides at the grain boundaries (2c in the chart above).

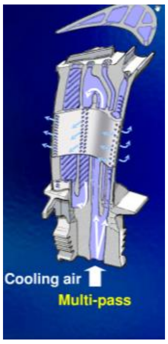
- For single crystal turbine blade: Single crystal have no (Ni)-(Ni) interfaces. Cannot have grain boundary sliding nor precipitates forming at grain boundaries. Many grain boundary strengthening solutes are removed giving higher solidus temperature. This allows for more effective heat treatments. (2b in the chart above).

4. How to achieve operating T higher than Ni alloy melting T

T/Tm of Ni alloy is higher than 1. How can we take the Ni alloy to its limit?

- In second generation cooling:

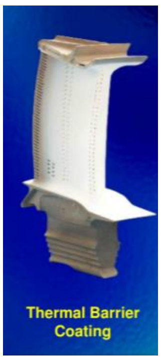
Cooling air from the compressor stage is fed through channels that eject the air around the blade. A film of coor air then envelopes the blade, allowing the blade to operate in even hotter environments.

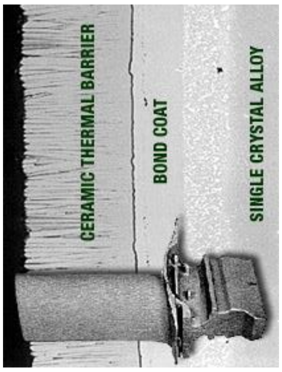


- In third generation cooling:

Cooling as per 2nd generation. Additionally, a coating of a ceramic (e.g. zirconium oxide) is applied, giving:

a. Protection from oxidation (the coatings are already oxides)
b. Thermal shielding (since oxides are poor conductors of heat, they provide insulation).





So the gas passing the turbine can be hotter than the melting point of the Ni alloy.